

Structure in the Recurrence

Conditioning Genomic Foundation Model Pretraining on 3D Chromatin Contacts

Project technical record

Compiled August 10, 2026

No experimental results appear in this document. No model has been trained. The abstract is a proposal abstract. Every number reported comes from a data-processing, parameter-accounting or unit-test script in the repository, never from a training run. The working rule for this project is that no figure reaches a draft unless it exists first in a logged output file.

Abstract

Genomic foundation models are pretrained on DNA sequence alone, yet much of the regulatory logic they are asked to predict is carried by the genome’s three-dimensional folding, which determines which enhancers can reach which promoters. Existing work combining sequence with chromatin contact data does so after the sequence representation is already fixed: CHROME trains an encoder on individual loci and only then adds a graph-attention layer over Hi-C contacts, and Evo2HiC distills a frozen Evo 2 into a compact encoder using Hi-C as a contrastive alignment target. In both, contact data shapes which learned features are combined, never which features are learned. We propose instead to place chromatin structure inside the pretraining computation. Contact-derived features modulate the state-transition dynamics of a bidirectional Mamba backbone, so Hi-C signal changes how far information propagates along the sequence rather than only what each position encodes. The mechanism adds 1.70% to parameter count over a matched sequence-only baseline. We evaluate whether a structure-shaped representation transfers—a question no sequence-to-structure model has been asked, as a forward-citation sweep over 318 papers citing Akita and Orca confirms—using long-range regulatory and variant-effect tasks together with a perturbation-sensitivity protocol on which Evo 2 has been reported to fail. Controls include structure shuffled at pretraining time and a distance-matched rewiring that tests whether any apparent structural signal is genomic distance in disguise.

Keywords: genomic foundation models; 3D genome organisation; Hi-C; topologically associating domains; state-space models; Mamba; self-supervised pretraining; CTCF; variant effect prediction; representation transfer.

Contents

1	The problem	2
2	Related work and the gap	2
2.1	The two nearest competitors	2
2.2	Two external findings that shaped the plan	3
2.3	The negative result the contribution rests on	3
3	Method	3
3.1	Baseline recurrence	3
3.2	The structural modification	3
3.3	Parameter accounting	4

4	Implementation	4
4.1	Scan backends	4
4.2	Dataset layer	5
5	Pilot data	5
5.1	Provenance	5
5.2	Measured properties	6
5.3	Structural features	6
5.4	Validation against independent tracks	6
6	Failure modes and controls	8
6.1	Shuffled-structure controls	8
6.2	The eight failure modes	8
6.3	Memory horizon, measured	9
7	Decisions of record	9
8	Change log	9
9	Open items	12
10	What happens next	13

1 The problem

An enhancer can sit 10^5 to 10^6 base pairs from the promoter it regulates. Nothing in the linear arrangement of the sequence explains how one reaches the other. The genome resolves this by folding: a ring-shaped cohesin complex extrudes a loop of DNA until it stalls at a pair of CTCF binding sites, and the resulting loop places the enhancer physically against the promoter.

The stalling condition matters for what follows. Cohesin halts only when the two CTCF motifs are *convergent*—pointing toward each other. Motif presence is insufficient; orientation decides. A sequence-only model sees the motif but receives no training signal that makes orientation pairing across a 500 kb gap consequential.

Three structures recur in the analysis. A *topologically associating domain* (TAD) is a region that contacts itself far more than its surroundings; enhancers act almost exclusively within their own TAD, so boundaries determine reachability. A *loop* is a point-to-point contact between two anchors. An *A/B compartment* is a coarse partition into open, active and closed, inactive regions.

2 Related work and the gap

Work	Seq.	Struct.	Structure enters at	Needed at inference
HiCFoundation (2026)	no	yes	pretraining	yes
Hi-Cformer (2025)	no	yes	pretraining	yes
Evo 2 (2026)	yes	no	—	no
Caduceus (2024)	yes	no	—	no
GraphReg (2022)	feat.	yes	supervised, downstream	yes
CHROME (2026)	yes	yes	stage 2, post-hoc	yes
Akita / Orca	yes	target	supervised target	no
Evo2HiC (2025)	yes	yes	contrastive distillation	no
This project	yes	yes	self-supervised pretraining	target: no

Table 1: Where 3D structure first influences the model. The bottom row is unoccupied in the current literature.

2.1 The two nearest competitors

CHROME (Ye et al., *Briefings in Bioinformatics* 27(4):bbag360, 2026) filters Hi-C against a self-avoiding polymer null model simulated over 5×10^5 chains, retaining 1.8–6% of contacts, then runs two graph-attention layers over signal-centred subgraphs with a 4 Mb receptive field at 5 kb bins. Training has two stages: the encoder is trained on centre nodes alone until validation saturates, and only afterwards are early layers frozen and the graph module attached. Supervision is 751 ENCODE assays. Structure never influences what the sequence encoder learns, and the contact graph is required at inference.

Evo2HiC (Fang et al., bioRxiv 10.1101/2025.11.18.689171) distils a frozen Evo 2 (7B) into a 3.6M-parameter CNN using two SigLIP contrastive objectives, the second aligning the student to Hi-C patch embeddings. Structure genuinely shapes a sequence encoder, but by aligning output embeddings, with no objective over nucleotides. A keyword count over the full preprint returns zero occurrences of *ClinVar*, *pathogenic*, *eQTL*, *variant*, *masked*, *MLM*, *Mamba* and *state space*.

Their third stated limitation is that they froze Evo 2 because fine-tuning it is computationally hard, and that they intend to develop strategies for adapting it with Hi-C data—this project’s question, named as work they have not done.

2.2 Two external findings that shaped the plan

Lee (arXiv:2604.07196, April 2026) probed Evo 2 (7B) on 231 TAD boundary regions and 120 convergent CTCF loops. Boundary deletions were penalised *less* than GC- and size-matched random controls (paired Wilcoxon $p = 0.405$); CTCF inversions and deletions likewise ($p = 0.021$, $p = 0.006$, in the wrong direction). Generated loops reached median enrichment 0.054 against a reference of 0.388. The paper concludes Evo 2 “has learned local CTCF grammar but misses higher-order 3D organization” and prescribes bidirectional architectures with explicit 3D contact inputs, citing Caduceus. This substantially defuses failure mode F3 (Section 6): if a 7B-parameter model with a 1M-token context cannot infer TAD-scale structure from sequence, a 7.7M-parameter model will not either.

DNALongBench (*Nature Communications* 16(1):10108, 2025) provides five long-range tasks including enhancer–target interaction, eQTL and contact-map prediction, already benchmarking Caduceus-Ph and Akita. Adopting it for evaluation removes the objection that tasks were chosen to flatter. Calibration: contact-map prediction is the hardest task in the suite, best reported correlation 0.233, by Akita.

2.3 The negative result the contribution rests on

A forward-citation sweep over 318 papers—118 citing Orca, 200 citing Akita—found no work that takes either model’s learned representations and evaluates them on a task other than predicting folding. Their variant applications all run the model twice, on reference and mutant sequence, and compare predicted contact maps. Internal embeddings are never extracted, frozen and probed, or fine-tuned into another head.

3 Method

3.1 Baseline recurrence

Per layer and direction, with channel index i and state index n :

$$(z, v) = \text{split}(W_{\text{in}} u_t) \quad (1)$$

$$x_t = \text{SiLU}(\text{Conv1d}(v)_t) \quad (2)$$

$$(\delta'_t, B_t, C_t) = \text{split}(W_x x_t) \quad (3)$$

$$\Delta_t = \text{softplus}(W_\delta \delta'_t + b_\delta) > 0 \quad (4)$$

$$A_{i,n} = -\exp(A_{i,n}^{\text{log}}) < 0 \quad (5)$$

$$\bar{A}_{t,i,n} = \exp(\Delta_{t,i} A_{i,n}) \in (0, 1) \quad (6)$$

$$h_{t,i,n} = \bar{A}_{t,i,n} h_{t-1,i,n} + \Delta_{t,i} B_{t,n} x_{t,i} \quad (7)$$

$$y_{t,i} = \sum_n C_{t,n} h_{t,i,n} + D_i x_{t,i} \quad (8)$$

3.2 The structural modification

Let $s_t \in \mathbb{R}^{d_s}$ be the structural context at position t , derived from Hi-C. Two terms are added:

$$\Delta_t = \text{softplus}(W_\delta \delta'_t + b_\delta + \textcolor{red}{W}_{\Delta s} s_t) \quad (9)$$

$$p_t = \text{softplus}(w_g^\top s_t + b_g) \geq 0 \quad (10)$$

$$\bar{A}_{t,i,n} = \exp(\Delta_{t,i} A_{i,n} - \textcolor{red}{p}_t) \quad (11)$$

The first retunes the memory length of each channel. The second is a non-negative damping that only shortens memory: at a TAD boundary the model can raise it and cut the state.

Why this is not an input feature. Appending s_t to the input would reach only $\bar{B}_t x_t$, the term governing how much of the current token enters. It cannot alter $\bar{A}_t$, which governs what survives. Because Δ enters $\bar{A}$ inside an exponential, structure here sets the effective memory horizon

$$\tau_{t,i} = \frac{-1}{\Delta_{t,i} A_{i,n} - p_t} \quad (\text{tokens}), \quad (12)$$

that is, how *far* information travels rather than what is encoded. This is the class of constraint that contrastive alignment on output embeddings cannot express, and it is the entire novelty claim.

Relationship to hard state resets. As $p_t \rightarrow \infty$, $\bar{A}_t \rightarrow 0$ and $h_t = \bar{B}_t x_t$: a hard reset. The mechanism is the soft, learned relaxation of TAD-conditioned state resetting, which is recorded as future work.

Initialisation. $W_{\Delta s} = 0$, $w_g = 0$, $b_g = -4$. The model is then numerically indistinguishable from the baseline at step zero, so any divergence is attributable to learned structural signal. Zero initialisation of $W_{\Delta s}$ does not block learning, since $\partial L / \partial W_{\Delta s} = (\partial L / \partial \Delta) s_t^\top$.

3.3 Parameter accounting

Computed by instantiating both models and summing parameter tensors (`scripts/param_accounting.py`, `scripts/test_model.py`).

Configuration	Total	Added	Δ
Sequence-only baseline	7,725,312	—	—
$d_s = 2$, with encoder	7,758,354	33,042	+0.428%
$d_s = 8$, with encoder	7,856,952	131,640	+1.704%
$d_s = 8$, no encoder (recommended)	7,856,672	131,360	+1.700%

Table 2: All configurations sit inside the 5% matched-parameter budget.

4 Implementation

4.1 Scan backends

The selective scan has two implementations, selected by device.

The reference implementation is a sequential loop over positions: correct, differentiable, and what the CPU unit tests run at small L . It is not viable at $L = 32,768$, which would require $32,768 \times 16 \times 2$ Python iterations per forward pass.

Production requires a fused kernel, and the two structural terms differ sharply in how they cooperate with the stock one. **The Δ bias is free:** `mamba_ssm`’s `selective_scan_fn` takes `delta` as an argument, so the bias is added before the call. **The permeability term p_t is not expressible** through that interface: it needs $\bar{A} = \exp(\Delta A - p)$, and folding p into Δ would require $\Delta' = \Delta + \log(g)/A_n$, which depends on the state index n . A single per-channel Δ cannot produce a decay uniform across n .

A custom Triton kernel was therefore written (`src/chromfm/scan_triton.py`), carrying p through both forward and backward. It uses one program per (batch, channel) pair, checkpoints

the state at chunk boundaries, and in the backward replays each chunk from its checkpoint before running the adjoint recurrence, keeping memory at $O(BD \frac{L}{\text{chunk}} N)$ rather than $O(BDLN)$.

The Triton kernel has never been executed. It was written on a CPU-only machine with no CUDA device and no Triton install. The recurrence and its adjoint were derived by hand and the code is syntactically complete, but nothing is numerically confirmed. `scripts/validate_kernel.py` must be run on the GPU box before any training. A scan with a subtly wrong gradient produces a smooth, plausible loss curve while training the wrong model.

Both arms must use the same backend, or the matched-compute comparison is confounded. The baseline therefore runs on this scan too, even though it never supplies p .

4.2 Dataset layer

Window		32,768 bp
Train stride		16,384 (50% overlap)
Evaluation stride		32,768 (no overlap)
Train windows		5,422
Validation windows	273	(chr9:120,029,184–128,974,848)
Test windows	242	(chr9:130,023,424–137,953,280)
Dropped	1,054 on N fraction, 799 on structural coverage, 126 buffer	

Table 3: Window index. Verified: zero training windows touch either held-out region; both evaluation splits are fully contained with no internal overlap.

Window length follows from the memory-horizon measurement rather than convention: the 16-layer relayed horizon is roughly 16,000 tokens, so 32,768 already exceeds what the stack can span.

Three policies are implemented in the dataset builder rather than in either training script, so the two arms cannot diverge on them:

- (i) **ϕ is interpolated, not stepped.** Linear interpolation between bin centres. Measured inside one window, a step function yields 7 distinct values per feature; interpolation yields 32,757. Seven values across a window is the precondition for failure mode F4.
- (ii) **N has its own token and a budget.** 12.00% of chr9 is unresolved; windows above 10% N are dropped.
- (iii) **Invalid bins are masked, never NaN.** A position is structurally valid only when both flanking bins are usable. Invalid positions receive $\phi = 0$, the standardised mean, plus a validity flag. A NaN reaching $W_{\Delta s} s_t$ would poison the entire recurrence.

Reverse-complement augmentation reverses and complements the tokens, then reverses ϕ and negates its two antisymmetric coordinates (`directionality_2Mb`, `upstream_mass_frac`).

5 Pilot data

5.1 Provenance

Source is 4D Nucleome experiment set **4DNES3JX38V5**: in situ Hi-C on GM12878, MboI with bio-dATP, Lieberman Aiden lab, published as Rao et al. 2014 (PMID 25497547), aligned to GRCh38. The multi-resolution contact matrix (4DNFIXP4QG5B) is 27.4 GB and is never downloaded; the pipeline reads a near-diagonal band directly over HTTP range requests in six and a half minutes. Boundary calls, insulation and compartment tracks are mirrored as independent validation targets, not as model inputs. Sequence is Ensembl release 113; annotation is GENCODE release 47.

5.2 Measured properties

chr9 length, from the contact matrix	138,394,717 bp
chr9 length, from the Ensembl FASTA	138,394,717 bp
Bins at 5 kb	27,679
Band entries retained	7,108,483
Band occupancy	0.6451
Bins with no balancing weight	19.31%
Unresolved sequence (N)	12.00%
Bins with a complete feature vector	21,519 (77.74%)

The two identical lengths are the first coordinate-alignment check. A second, stronger one emerged unplanned: the pipeline returned nothing for chr9 between 48 and 58 Mb. That is the centromere and the 9q12 heterochromatic block, which GRCh38 represents as a gap. The sequence file is 100% unresolved from approximately 46 to 60 Mb, and the contact matrix has no usable bins from 43.27 to 60.52 Mb. Two files from different institutions placing the same feature at the same coordinates is stronger evidence than a matching chromosome length.

5.3 Structural features

Index	Feature	Window	Under reverse complement
0–2	insulation score	100, 250, 500 kb	symmetric
3	directionality index	2 Mb	antisymmetric
4	log contact density	2 Mb	symmetric
5	upstream mass fraction	2 Mb	antisymmetric
6	short-to-long range ratio	100 kb / 2 Mb	symmetric
7	A/B compartment eigenvector	250 kb, GC-oriented	symmetric

Table 4: The symmetry column is load-bearing: failure mode F7 is the case where features 3 and 5 cancel between the forward and reverse passes.

5.4 Validation against independent tracks

Our feature	Against 4DN’s own track	Pearson r
Insulation, 100 kb window	4DNFIBMOGOZC	+0.9969
Insulation, 250 kb window	4DNFIBMOGOZC	+0.7802
Insulation, 500 kb window	4DNFIBMOGOZC	+0.4895
A/B compartment eigenvector	4DNFILYQ1PAY	+0.9759

Table 5: The 100 kb window reproduces 4DN’s track almost exactly, identifying their diamond window and confirming the implementation. Declining correlation at larger windows is the expected consequence of comparing different window sizes against a fixed reference.

A TAD and a loop are both visible in the processed data. The loop lies at chr9:136,485,000 paired with 136,625,000, 140 kb apart, at 7.5 times its distance-matched expectation, with six CTCF ChIA-PET read pairs from an independent assay at the same anchors. Its upstream anchor falls on *NOTCH1* (chr9:136,494,433–136,546,048).

Phase 1 step 3 — pilot pipeline visual validation (GM12878 in situ Hi-C, Rao et al. 2014, via 4DN, GRCh38)

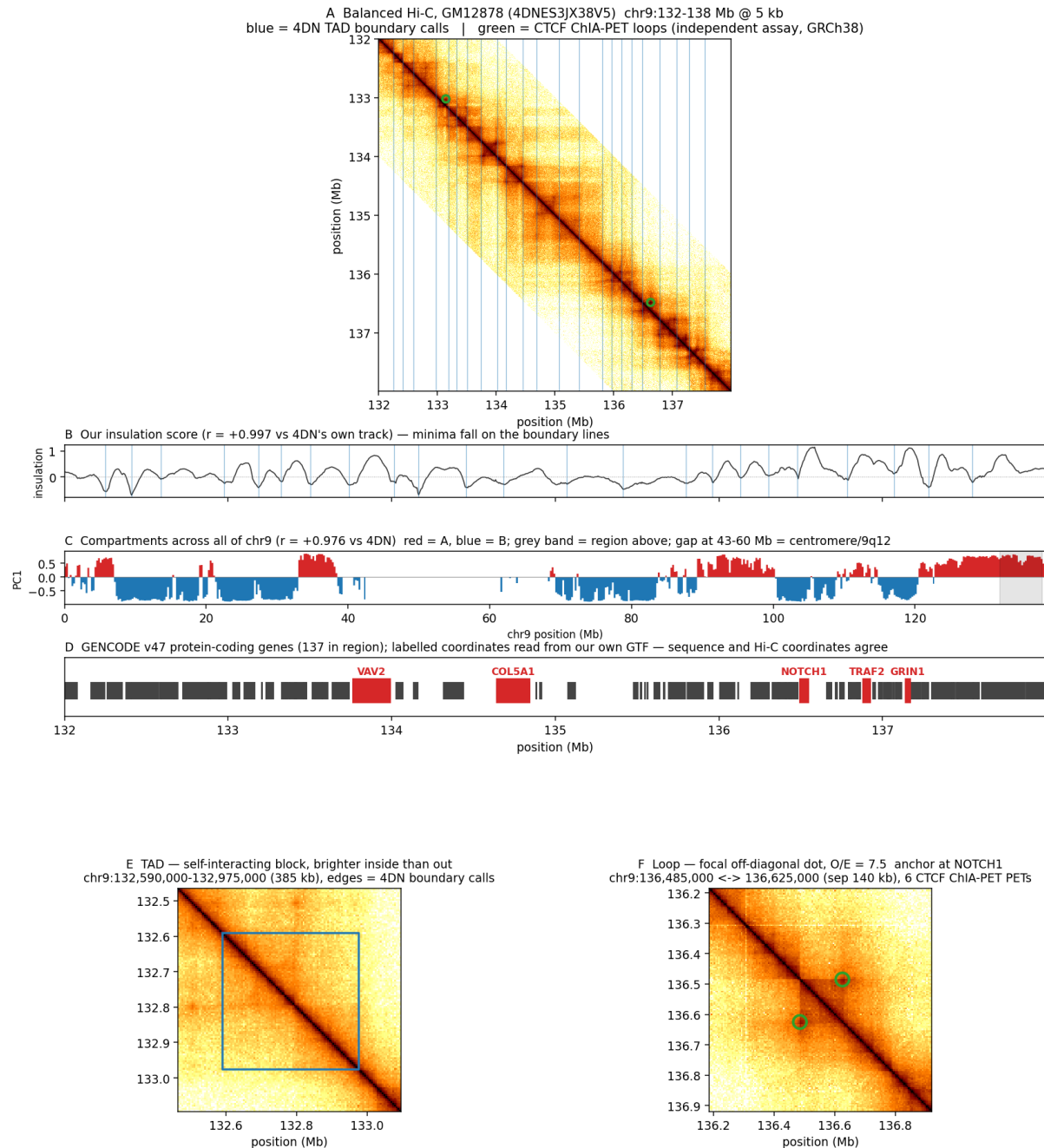


Figure 1: Pilot pipeline validation. (A) Balanced contacts with 4DN boundary calls in blue and CTCF ChIA-PET loops in green. (B) Our insulation score. (C) Compartments across all of chr9; the blank interval at 43–60 Mb is the centromere. (D) GENCODE genes with coordinates read from our own annotation. (E) A 385 kb TAD. (F) The NOTCH1-anchored loop.

6 Failure modes and controls

6.1 Shuffled-structure controls

ID	Control	Destroys	Rules out
S0	zero the signal	everything	establishes the sequence-only floor at identical parameter count
S1	permute across bins	alignment and autocorrelation	primary reliance probe
S2	circular shift, 10 Mb	alignment only	“any smooth auxiliary channel would do”
S3	distance-matched rewire	locus-specific structure	“structure is genomic distance in disguise”

Table 6: S3 is the control that can end the project.

Contact probability falls smoothly with genomic distance. If a model performs as well on distance-matched fake structure as on real structure, what it learned is a re-parameterised positional prior and the central claim is false however good the headline number looks.

The gate passes only if degradation under shuffling is positive with a 95% bootstrap confidence interval excluding zero *and* is at least twice the across-seed standard deviation of the real-structure runs. Significance alone is insufficient; the effect-size floor prevents a p -value standing in for an effect.

6.2 The eight failure modes

Failure	Symptom	Status
F1 bias absorbed into b_δ	trains to baseline loss	highest probability; mitigated by genome-wide standardisation
F2 softplus saturation	structural weights never move	observed ; b_g changed $-8 \rightarrow -4$
F3 structure inferable from sequence	structural input redundant	largely defused by Lee’s Evo 2 probe
F4 bin longer than memory horizon	trains to baseline loss	precondition confirmed ; mitigated by interpolation
F5 permeability gate stays at zero	half the mechanism inert	monitor the distribution of p
F6 gradient starvation	structural pathway never learns	1.7% of parameters behind two nonlinearities
F7 forward and reverse cancel	trains to baseline loss	two of eight features flip sign under RC
F8 structural encoder collapses	reduces to F1	eliminated if the encoder is removed

F1, F4 and F7 produce the same symptom: the model trains to baseline performance and raises no error. Four diagnostics therefore run continuously—the variance the structural pathway contributes to the timescale, the distribution of the permeability penalty, the growth of the

structural weight norm, and a probe asking whether the baseline model’s hidden states already predict the structural signal.

6.3 Memory horizon, measured

At the reference initialisation, horizons run from 0.6 tokens at the fast end to 994 at the slowest, median 14.9. A 5 kb bin is 5,000 tokens, so no channel-state pair spans a single bin; $\tau_{\max}/\text{bin} = 0.199$. Sixteen layers relaying extends reach to roughly 15,900 tokens, still short of a 385 kb TAD.

Δ is learned and can grow, so this is not proof of failure. It constrains the sequence-only baseline exactly as much as the structural model, which raises a question worth settling before spending GPU time: can a model of this size represent TAD-scale dependencies at all? **Phase 3 must therefore report the trained memory horizon, and that measurement gates Phase 4.**

7 Decisions of record

	Decision	Choice	Reasoning
1	Mechanism	structural bias in the recurrence	unreachable by contrastive alignment on output embeddings; contains hard state resets as its limit; cheap to falsify
2	Reverse complement	Caduceus-PH (augmentation)	equivariance is orthogonal to the hypothesis; coupling them makes a null result uninterpretable
3	Hi-C resolution	5 kb, 1 kb as ablation	comparability with CHROME; both levels are in the same file
4	Cross-species scope	out for this paper	Evo2HiC uses 177 species, HiC-Foundation 316; entering that lane splits compute against better-resourced groups
5	Permeability term	keep , via custom kernel	retains the soft-reset semantics and the link to future work

8 Change log

Every substantive change to the project is recorded here with its reason.

2026-08-10

First session on the L40S box. Environment built, pipeline reproduced from a clean tree, custom kernel executed for the first time. No model trained.

- **Environment.** torch 2.6.0+cu124, triton 3.2.0, Python 3.10, venv 3d-gen/ (added to .gitignore). torch.cuda.device_count() = 2; both devices report NVIDIA L40S, compute capability 8.9, 44.4 GiB.
- **NVML is broken on this box and nvidia-smi does not run.** The loaded kernel module is 580.126.09 while the userspace libraries are 580.173.02. CUDA is unaffected—cuInit

returns 0, both devices enumerate and compute—but NVML-based utilisation and temperature logging is unavailable until the modules are reloaded, which needs root. Recorded because the run config is supposed to carry the driver version, and the number `nvidia-smi` would have reported cannot be obtained here.

- **Pipeline reproduced exactly from a clean tree.** `phase1_acquire.py` assembled **7,108,483** unique upper-triangle band entries with all three auxiliary md5s verified; `phase1_features.py` reproduced insulation $r = +0.9969$ ($n = 21,740$) and compartments $r = +0.9759$ ($n = 22,250$) against 4DN’s own tracks, with 21,519/27,679 bins carrying a complete ϕ ; `phase1_dataset.py` built **5,422 / 273 / 242** train/val/test windows with drop counts 1,054 (N fraction), 799 (structural coverage), 126 (buffer), 2 (short). Every one of these matches `docs/data_card.md` exactly, so the data layer is confirmed reproducible on a second machine rather than only on the machine that built it.
- **data/pilot_manifest.json now differs from the committed copy, and none of the differences are content.** The committed manifest was written on Windows. Rebuilding on Linux changed (i) path separators, `data\raw\` to `data/raw/`; (ii) the byte size of the two text files, because the Windows copies had CRLF line endings —`chr9.fa` shrank by exactly 2,306,580 bytes against 2,306,580 lines, and the GTF by exactly 169,673 bytes against 169,673 lines, one byte per line in each case; and (iii) the md5 of the two `.npz` files, whose byte sizes are nevertheless *identical*, because a zip central directory records the creating platform—`create_system` is 3 (Unix) here and 0 (Windows) there. The md5s of the three files downloaded from 4DN, which are the ones checked against the 4DN API, are unchanged. Band contents re-verified directly: 7,108,483 entries, 0 non-finite, min 4.25×10^{-6} / median 3.10×10^{-4} / max 0.942, 27,679 bins, 5,345 (19.31%) without a balancing weight—every figure matching `docs/data_card.md` §4. Whether to commit the Linux manifest is a judgement call left to the PI; a manifest whose md5s are platform-dependent cannot serve as a cross-platform integrity check for the derived files, only for the downloaded ones.
- **test_model.py: 16/16 passed** on this box, unchanged.
- **Triton kernel executed for the first time and validated: 34/34 checks passed** (`validate_kernel.py`, exit 0) across no- p , with- p , multi-chunk and ragged-final-chunk configurations, including the assertion that $\partial L / \partial p \neq 0$.
- **Three kernel defects fixed, all in the Triton lowering rather than the mathematics.** The hand-derived adjoint was correct as written. (i) `CHUNK` was a module-level global, which Triton refuses to read from a `@jit` function; it is now a `tl.constexpr` kernel parameter. (ii) The chunk walk used `range(NCHUNK-1,-1,-1)`; Triton lowers `range` to `scf.for`, which requires a non-negative step, so both reversed loops now count up and invert the index by hand. (iii) The inner loops used `tl.minimum(CHUNK, L-t0)` as a trip count; they now always run `CHUNK` times and predicate on $t < L$.
- **The backward is not bitwise reproducible.** `dBmat`, `dCmat` and `dp` are accumulated with `tl.atomic.add` across the d_{inner} programs, so their summation order varies between runs. Measured relative spread over five repeats at $B = 2$, $D = 512$, $L = 512$: 1.0×10^{-6} , 9.6×10^{-7} and 8.9×10^{-7} respectively; `du`, `ddelta`, `dA` and `dDskip` were bitwise identical. This is fp32 rounding noise, not a defect, but it means seed-identical runs reproduce to fp32 tolerance and not to the bit. The paper must not claim bit-exact reproducibility.
- **Kernel checked beyond the supplied harness**, which runs at $D = 64$. Added checks at the real width $d_{\text{inner}} = 512$, at 64 chunks, and at $L = 1$ and $L = 65$; all agree with the reference. Width matters because the cross-channel atomics only contend at width.
- **Scan checkpoint memory measured**, since it bounds the Phase 3 batch size: 16.0 MiB per sample per scan at $d_{\text{inner}} = 512$, $d_{\text{state}} = 16$, $L = 32,768$, which is 0.50 GiB per sample across 16 layers and 2 directions, or 4.0 GiB at batch 8 before any other activation.

- **TeX Live installed user-local** (TinyTeX). The box had no LaTeX engine and no root, so `build_project_doc.py` could not run at all.

2026-08-10 (second entry) — Phase 3 training harness

`scripts/train.py` written. Results are recorded in a later entry; this one covers the harness and the platform findings only.

- **NCCL cannot be used on this box.** It calls `nvmlInit_v2()` during topology discovery, which fails for the same driver mismatch that breaks `nvidia-smi`. `init_process_group("nccl")` is lazy and succeeds anyway, so the failure surfaces at the first collective inside DDP's constructor. The script therefore probes NVML directly and falls back to **gloo**. Measured cost of the fallback: none detectable—two GPUs give $2.00\times$ the single-GPU throughput (0.756 windows/s on one L40S, 1.51 windows/s on two).
- **PyTorch's CUDA caching allocator also calls NVML** when it reports memory pressure, so an out-of-memory condition surfaces as `INTERNAL ASSERT FAILED ... nvmlInit_v2_` rather than as `CUDA out of memory`. Anyone reading that traceback later should read it as OOM.
- **Kernel made $1.42\times$ faster, and re-validated.** The scan's state vector is $d_{\text{state}} = 16$ elements wide, so Triton's default `num_warps=4` gave each program 128 lanes for 16 lanes of work. Setting `num_warps=1` took the real config from 7.55 to 5.30 s/step. Per-step losses were identical to the default across the benchmark, and `validate_kernel.py` was re-run afterwards: 34/34.
- **Gradient checkpointing implemented and then measured not to help.** The hypothesis was that the scan is latency-bound and a bigger batch bought with recompute would be nearly free. Batch scaling turned out to be linear—the GPU is already saturated at batch 2—so checkpointing costs its recompute undiluted: 1.32 s/window at batch 2 without it against 1.49 s/window at batch 8 with it. The code is retained behind `--grad-checkpoint`, off by default, as the lever to reach for if d_{model} or the window grows.
- **fp32 throughout, no autocast**, because the scan was validated in fp32 only. Training the experiment in bf16 would put it back on an unvalidated numeric path through a hand-written recurrence.
- **The baseline runs the custom Triton scan** even though it never supplies p , so the matched-compute claim against the structural arm is real rather than nominal.
- **N is excluded from masking and from the loss.** 12.00% of chr9 is N and a window may carry up to 10%. Including N would deflate bits/nucleotide by an amount that varies with each window's N content. The reported metric is therefore bits per *resolved* nucleotide, and its uniform-random floor is exactly 2.000 bits.
- **τ is measured two ways and they are not interchangeable.** `BiMambaLM.tau_stats()` derives τ from `dt_proj.bias` alone. That is a fair proxy at initialisation, where `dt_proj.weight` is small, but after training Δ is dominated by the input-dependent term $W_{dt}\delta'_t$, and a bias-only τ can be badly wrong. The script logs `tau_stats()` because the runbook asks for it, and separately measures *empirical* τ from the Δ actually produced on validation batches. **The F4 gate must be read off the empirical numbers.**
- **num_workers defaults to 0.** `WindowDataset` owns an `np.random.default_rng` for reverse-complement augmentation; under multiple loader workers each worker copies it, so the augmentation stream would depend on how the sampler sharded and seeds would stop being reproducible.

2026-08-07

- **Phase 0 completed.** Eleven papers mapped. Two competitors found that were not on the original reading list: Evo2HiC, which is closer to the thesis than CHROME, and Lee’s Evo 2 probe, which supports it.
- **Differentiator retired.** “Structure at training, sequence-only at inference” was dropped as a claim against Evo2HiC, whose DNA-only encoder has the same property. It still holds against CHROME.
- **Forward-citation sweep.** 318 papers checked; no work probes Akita or Orca representations on a non-folding task. The transfer gap stands.
- **Mechanism (a) selected;** (b) and (c) documented but not pursued.
- **Memory horizon measured.** $\tau_{\max} = 994$ tokens against a 5,000-token bin. Consequence: ϕ interpolation became mandatory, and a trained- τ measurement was added as a Phase 4 gate.
- **Pilot acquired and validated.** Insulation $r = +0.9969$ and compartments $r = +0.9759$ against independent 4DN tracks. A TAD and a ChIA-PET-corroborated loop confirmed visually.
- **Three assembly traps caught.** A 1,425 kb “loop” that was sparse-corner noise; an hg19 *ABL1* coordinate used to choose a GRCh38 region; and the standard GEO loop list for this dataset, which is hg19—detected because its largest chr9 anchor exceeds the length of GRCh38 chr9.
- **Dataset layer built.** Splits verified exact after a first version allowed held-out windows to straddle their region edge.
- **Model implemented; two bugs found by unit tests.** The reverse-complement sign flip was being applied to the encoded signal rather than to raw ϕ , where the encoder has already mixed all eight coordinates, making the operation meaningless. Separately, A was broadcast by trailing-axis alignment, which matches the state dimension against sequence length and only agrees when $L = d_{\text{inner}}$; the first test passed only because it used $L = 64$ with $d_{\text{inner}} = 64$.
- **b_g changed from -8 to -4 .** The gradient through softplus is $\sigma(b_g)$, so -8 attenuated the gate’s learning signal roughly 3000-fold—failure mode F2, self-inflicted by our own initialisation. Measured effect on the gate gradient: $2.06 \times 10^{-7} \rightarrow 7.69 \times 10^{-6}$.
- **Encoder bottleneck identified.** $\partial L / \partial s = W_{\Delta s}^\top \partial L / \partial \Delta$ with $W_{\Delta s}$ zero-initialised, so the encoder receives exactly zero gradient on step zero and roughly 10^{-9} afterwards against 10^{-4} for the matrix in front of it. Recommendation: remove the encoder and feed the eight standardised features directly at $d_s = 8$, costing +1.700% parameters and eliminating F8.
- **Triton kernel written** for the permeability term, with a GPU validation harness. Unverified until executed.

9 Open items

- **Decide whether bit-exact reproducibility is required.** If it is, the atomics in the backward must be replaced with a deterministic reduction, at a performance cost. If it is not—the reasonable reading, given that the Phase 5 gate is across-seed variance rather than a single number—say so explicitly, because the spread is measured and small.
- **Restore nvidia-smi** by reloading the NVIDIA kernel modules, or accept that no driver-level utilisation figure can be logged alongside the runs.
- **Confirm the encoder removal** ($d_s = 8$, no encoder), which changes parameter figures recorded in three documents.
- **Read CHROME line by line.** The summary here came from automated extraction of

the PMC full text.

- **Obtain Orca’s Results section**, paywalled at *Nature Genetics* with no PubMed Central deposit. The claim that its representations were never evaluated on a non-folding task rests on the abstract, all ten extended-data captions and the section headings, not the body.
- **Re-check Evo2HiC** for a revision adding variant evaluation before Phase 4 commits GPU time; that would close the one contribution nothing currently contests.
- **Re-check Lee’s preprint** for peer-review status before it carries weight in a methods section.

10 What happens next

Phase 3 trains the sequence-only baseline, logging every hyperparameter, seed and metric before any structural model exists, so no flattering comparison can be selected after the fact. That run must also report the trained memory horizon, which decides whether a model of this size can represent TAD-scale dependencies at all. If it cannot, the mechanism is re-scoped to sub-TAD structure or the model is enlarged, and that decision is taken before Phase 4 rather than after.

Every measured number in this document traces to a named script: `phase1_acquire.py`, `phase1_features.py`, `phase1_validate_visual.py`, `phase1_dataset.py`, `param_accounting.py`, `f4_memory_horizon.py`, `test_model.py`. Literature claims are sourced in `docs/related_work.md`; the mechanism is specified in `docs/architecture_spec.md`; data provenance is in `docs/data_card.md`.